CHAPTER 3

Three Strategies and the Probability Simplex

Three population proportions require two independent coordinates. Their state space is a triangle, and a trajectory can turn as it moves through that triangle. Cyclic interactions supply a concrete planar system in which trajectories can approach equilibrium, circulate periodically, or move toward the boundary.

The interval of the preceding chapters recorded a single proportion; its direction of motion was determined by a sign. On the probability triangle, a velocity has two independent components. The possibility of turning makes planar geometry essential. The edges of the triangle still contain the familiar two-type models, while the interior permits genuinely three-type behavior.

3.1 A triangle of normalized states

Let p_A, p_B, p_C be the proportions of three types. They satisfy

$$p_A + p_B + p_C = 1, \qquad p_A, p_B, p_C \geq 0.$$

The set of such vectors is the *standard 2-simplex*:

$$\Delta^2 = \{(p_A, p_B, p_C) \in \mathbb{R}^3 : p_i \geq 0,\ p_A + p_B + p_C = 1\}.$$

It is the triangular region cut from the first octant by the normalization plane. The vertices $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$ are pure states. An edge consists of mixtures in which one type is absent. All three types are present in the relative interior.

For a flat equilateral drawing, choose vertices $A = (0,0)$, $B = (1,0)$, and $C = (1/2, \sqrt{3}/2)$. The displayed position of a state is

$$r(\mathbf{p}) = p_A A + p_B B + p_C C = \left(p_B + \frac{1}{2} p_C, \frac{\sqrt{3}}{2} p_C \right). \tag{3.1}$$

The probabilities are *barycentric coordinates*: the state is their weighted average of the three vertices. A point halfway along the AB edge has state $(1/2, 1/2, 0)$.

One can instead use Cartesian coordinates $x = p_A$, $y = p_B$, and $p_C = 1 - x - y$. Their domain is the right triangle $x \geq 0$, $y \geq 0$, $x + y \leq 1$. These two drawings give different linear pictures of the same states.

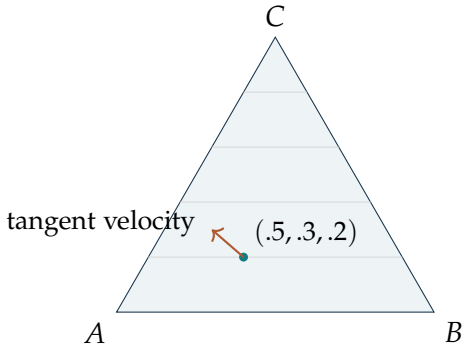


Figure 3.1. A three-type population state and a tangent velocity in an equilateral simplex drawing. The planar arrow is the image of a three-component velocity whose components sum to zero.

3.2 The general replicator system

Let $N_i' = f_i N_i$ for $i = A, B, C$, and write $N = \sum_i N_i$ and $p_i = N_i / N$. Then

$$N' = N \sum_i p_i f_i.$$

The quantity

$$\bar{f} = \sum_i p_i f_i$$

is the *mean fitness*. Differentiating p_i gives

$$\boxed{p_i' = p_i (f_i - \bar{f}).} \tag{3.2}$$

The same derivation applies to any finite number n of types. Each type changes according to its advantage over the current population average.

Example 3.1 (Computing a velocity)**.** At $\mathbf{p} = (0.2, 0.3, 0.5)$ suppose $\mathbf{f} = (3, 1, 2)$. Find the mean fitness and the replicator velocity.

Solution. The mean is $0.2(3) + 0.3(1) + 0.5(2) = 1.9$. Thus

$$\mathbf{p}' = (0.2(1.1), 0.3(-0.9), 0.5(0.1)) = (0.22, -0.27, 0.05).$$

Types A and C gain share. The components sum to zero, so the increases exactly balance B's decrease.

For n types, the closed state space is Δ^{n-1}, of dimension $n - 1$. The constraint removes one independent coordinate. A differentiable curve in its normalization plane has tangent velocity satisfying

$$\sum_i p_i' = 0. \tag{3.3}$$

For the replicator field this follows algebraically:

$$\sum_i p_i(f_i - \bar{f}) = \bar{f} - \bar{f}\sum_i p_i = 0.$$

3.3 Invariant faces and positive solutions

If $p_k = 0$, then the kth replicator component is zero. With smooth fitness functions, uniqueness therefore makes every face of the simplex invariant. A population that begins with a type absent retains that absence under the model.

An explicit identity describes positivity. Along a solution with $p_i(0) > 0$,

$$p_i(t) = p_i(0) \exp\left(\int_0^t [f_i(\mathbf{p}(\tau)) - \bar{f}(\mathbf{p}(\tau))]\,\mathrm{d}\tau\right). \tag{3.4}$$

If the fitness functions are smooth on a neighborhood of the compact simplex, the integrand is bounded on every finite time interval. The exponential is then positive. A type present initially remains present at finite times, although its proportion can tend to zero as time increases without bound.

Example 3.2 (An edge carries a two-strategy equation)**.** Consider the matrix

$$M_s = \begin{pmatrix} 0 & 1 & -s \\ -s & 0 & 1 \\ 1 & -s & 0 \end{pmatrix}, \qquad s > 0.$$

Reduce its dynamics to the AB edge, where $p_C = 0$.

Solution. Set $p_A = x$ and $p_B = 1 - x$. Then $f_A = 1 - x$ and $f_B = -sx$. The edge equation is

$$x' = x(1-x)\big[1 + (s-1)x\big].$$

The bracket is positive on $[0,1]$ because its endpoint values are 1 and s. A therefore wins along the AB edge. The absent C type stays absent, even though it would interact with A if it were introduced.

Interior and edge trajectories can consequently have different limiting behavior. A small positive introduction of a missing type changes the state from one invariant face to a larger state space. This distinction is essential when interpreting the triangle's boundary.

3.4 Cyclic competition

For a payoff matrix M, random mixing gives

$$\mathbf{f} = M\mathbf{p}, \qquad \bar{f} = \mathbf{p}^T M\mathbf{p}.$$

The cyclic family M_s above assigns payoff $+1$ to a win, $-s$ to a loss, and zero to an encounter with the same type. A outperforms B, B outperforms C, and C outperforms A. Its fitnesses are

$$f_A = p_B - sp_C, \quad f_B = p_C - sp_A, \quad f_C = p_A - sp_B. \tag{3.5}$$

At the uniform state $\mathbf{p}^* = (1/3, 1/3, 1/3)$, all three fitnesses equal $(1-s)/3$. Hence $\mathbf{p}^*$ is an equilibrium for every $s > 0$.

Example 3.3 (A first direction of cyclic motion)**.** For $s = 1$ and $\mathbf{p} = (1/2, 1/3, 1/6)$, compute the fitnesses and the velocity.

Solution. The fitness vector is $(1/6, -1/3, 1/6)$. Its mean is zero, so

$$\mathbf{p}' = \left(\frac{1}{12}, -\frac{1}{9}, \frac{1}{36}\right).$$

The sum is zero. A and C increase at this instant, while B decreases. The identities of the increasing types change as the state moves, producing circulation.

When $s = 1$, the matrix is antisymmetric: $M_1^T = -M_1$. For every vector $\mathbf{p}$, the scalar $\mathbf{p}^T M_1 \mathbf{p}$ equals its transpose, which is also $-\mathbf{p}^T M_1 \mathbf{p}$. Thus the mean fitness is identically zero. Individual fitnesses can differ from zero and drive motion.

3.5 A product that detects inward and outward motion

The pictures of cyclic motion suggest three different tendencies: inward spiraling, closed circulation, and outward spiraling. To distinguish them beyond the linear approximation near the center, we seek a scalar quantity whose change has a definite sign throughout the interior. The product of the three probabilities serves this purpose. For an interior state, define

$$P = p_A p_B p_C, \qquad Q = p_A p_B + p_B p_C + p_C p_A.$$

Differentiating the logarithm of the product gives

$$\frac{d}{dt} \ln P = \sum_i \frac{p_i'}{p_i} = \sum_i f_i - 3\bar{f}.$$

The cyclic fitnesses satisfy $\sum_i f_i = 1 - s$, while

$$\bar{f} = (1-s)Q.$$

Therefore

$$\boxed{\frac{d}{dt} \ln P = (1-s)(1-3Q).} \tag{3.6}$$

The second factor is nonnegative. Indeed,

$$1 - 3Q = \frac{1}{2}\left[(p_A - p_B)^2 + (p_B - p_C)^2 + (p_C - p_A)^2\right],$$

and it vanishes precisely at the uniform state. This identity determines the direction of product change throughout the interior.

If $s < 1$, the product increases at every nonstationary interior point. Its maximum is $1/27$, attained at the center. If $s > 1$, it decreases. If $s = 1$, it is conserved.

Proposition 3.1 (Balanced cyclic orbits)**.** *For $s = 1$, every nonstationary interior trajectory lies on a periodic orbit surrounding $\mathbf{p}^*$.*

Proof. The function $\ln P = \ln p_A + \ln p_B + \ln p_C$ is strictly concave in the simplex interior, tends to $-\infty$ at the boundary, and has a unique critical point at the center. For $0 < c < 1/27$, the level $P = c$ is a smooth closed curve surrounding the center. Conservation makes the vector field tangent to this curve. An interior equilibrium requires equal fitnesses, which in this balanced model forces all three probabilities to agree. Thus the field is nonzero on $P = c$. A continuous nonzero velocity on a compact smooth circle carries the state around the circle in finite time and repeats. □

The monotonicity also explains the other two regimes globally. For $s < 1$, a trajectory remains in the compact interior set $P \geq P(0) > 0$. Any limiting invariant set must have constant limiting product, hence must lie where (3.6) vanishes. That is the center. For $s > 1$, a noncentral trajectory has decreasing product. If its limit were positive, the same compactness argument would force convergence to the center, whose product is larger than the initial product. This is impossible; therefore $P \to 0$, and the trajectory approaches the boundary as a set.

The last conclusion allows prolonged visits near different vertices. It does not specify convergence to one pure state. The directions along the edges form a cycle, so the currently predominant type can continue changing.

3.6 Planar linearization at the center

Use independent coordinates $x = p_A$ and $y = p_B$, so $p_C = 1 - x - y$. For a planar system $(x', y') = (F, G)$, the Jacobian at an equilibrium is

$$J = \begin{pmatrix} \partial_x F & \partial_y F \\ \partial_x G & \partial_y G \end{pmatrix}.$$

For the cyclic family, substituting the fitnesses and evaluating at $(1/3, 1/3)$ yields

$$J_s = \frac{1}{3}\begin{pmatrix} s & s+1 \\ -(s+1) & -1 \end{pmatrix}. \tag{3.7}$$

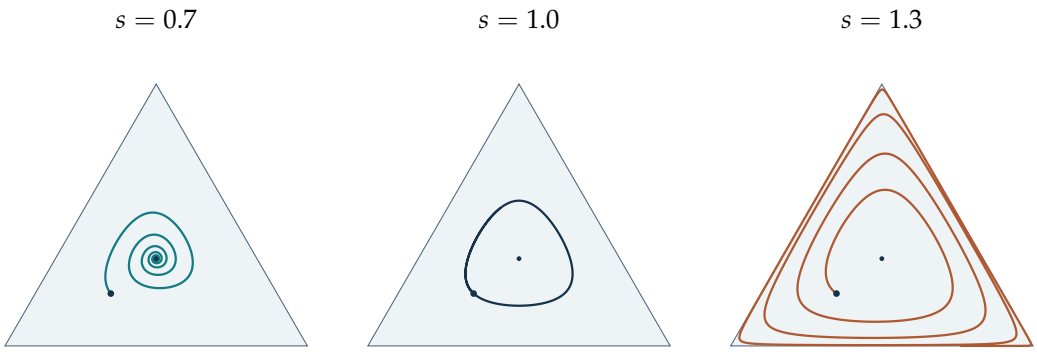


Figure 3.2. Cyclic replicator trajectories from the same initial law $(.55, .25, .20)$. Left: attraction toward the center. Middle: a closed product level. Right: outward motion toward the boundary. Curves are numerically integrated from the stated vector fields.

For instance, at the center $f_A - \bar{f} = 0$, so differentiating $x(f_A - \bar{f})$ leaves $x\,\partial(f_A - \bar{f})$. Carrying this out for the four entries gives the displayed matrix.

Its trace and determinant are

$$\operatorname{tr} J_s = \frac{s-1}{3}, \qquad \det J_s = \frac{s^2+s+1}{9}.$$

The characteristic equation gives

$$\lambda_\pm = \frac{s-1}{6} \pm i\frac{\sqrt{3}(s+1)}{6}. \tag{3.8}$$

For $s > 0$, the imaginary parts are nonzero. The real parts are negative for $s < 1$ and positive for $s > 1$, giving a locally attracting spiral and repelling spiral, respectively. At $s = 1$, linearization produces a center. The conserved product establishes the periodic behavior of the nonlinear system in that case.

Example 3.4 (Damping and oscillation)**.** For $s = 0.7$, estimate the decay rate and angular frequency near the center.

Solution. The eigenvalues have real part -0.05 and imaginary parts approximately ± 0.49075. In the linearized system a perturbation has an exponentially decaying envelope $e^{-0.05t}$ while it rotates with angular frequency about 0.49075. The corresponding small-oscillation period is approximately $2\pi/0.49075 = 12.80$.

3.7 Finding equilibria and testing invasion

At a replicator equilibrium, every present type has fitness equal to the population mean. Types with zero proportion impose no stationarity condition on their own fitness values, because their equations already vanish through the factor p_i.

For an interior matrix equilibrium, all types are present. The condition can therefore be written

$$M\mathbf{p} = c\mathbf{1}, \qquad \mathbf{1}^T\mathbf{p} = 1,$$

where c is the common fitness. This is a linear system for the probabilities together with c. A solution represents an interior equilibrium when all its probability coordinates are positive. Singular systems can have no solution or a family of solutions, so positivity and uniqueness must be checked after solving the equations.

For the cyclic family, subtracting pairs of the equal-fitness equations gives the uniform solution. One can also see uniqueness by restricting M_s to the tangent plane: the corresponding linear equal-fitness system has nonzero determinant for $s > 0$. The three vertices remain equilibria independently of that interior calculation.

Stability on an invariant edge concerns perturbations within that edge. A missing type can introduce another direction of perturbation. If a face equilibrium is q and type k is absent, its equation near the face has leading term

$$p_k' \approx p_k\big(f_k(q) - \bar{f}(q)\big).$$

The difference is the type's invasion rate at the resident equilibrium. A positive value makes a small introduction grow to first order, even if the resident state attracts along its own face.

Example 3.5 (An edge attractor with an invading third type)**.** Consider

$$M = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 3 & 3 & 0 \end{pmatrix}.$$

Analyze the AB equilibrium $q = (1/2, 1/2, 0)$ along the edge and under introduction of C.

Solution. On the edge, $f_A = 2(1-x)$ and $f_B = 2x$, giving $x' = x(1-x)(2-4x)$. The equilibrium $x = 1/2$ attracts, with derivative -1.

At the same state in the full model, the fitnesses are $(1, 1, 3)$ and mean fitness is 1. A rare C type has invasion rate $3 - 1 = 2$, so its initial growth is approximately $p_C' = 2p_C$. The equilibrium attracts along the edge and repels in the direction introducing C. It is unstable in the full simplex.

At a pure resident state e_j, the invasion rate of type $i \neq j$ is particularly simple: $m_{ij} - m_{jj}$. These are the diagonal entries of the tangent linearization in the rare-type coordinates. In balanced cyclic competition, a pure A population resists rare B with rate -1, while rare C invades with rate $+1$. Each vertex similarly has one attracting

and one repelling edge direction. These local directions organize the circulation around the boundary.

Thus the triangle carries several related stability questions: behavior within each invariant edge, invasion of the missing type, and motion near an interior equilibrium. Their coordinate calculations are elementary, but they describe distinct allowable perturbations of the population state.

3.8 Distributions, support, and state-space dimension

The *support* of a probability vector is the set of indices with positive coordinates. Smooth replicator dynamics preserve support at finite times. A face of the simplex is therefore both a geometric object and the state space of a population restricted to its initially present types.

The same normalized vector also describes the result of sampling one individual. The triangle is a space of categorical distributions, and the replicator field supplies one rule for moving through it. A distribution can be studied independently of that rule, which permits the probability and information constructions developed next.

3.9 Exercises

3.1 Plot $(0.2, 0.5, 0.3)$ using (3.1). Identify the absent type on each edge of the triangle.

3.2 Show that the three-category simplex has two independent coordinates and that its tangent velocities have components summing to zero.

3.3 For $\mathbf{p} = (1/2, 1/4, 1/4)$ and $\mathbf{f} = (2, 0, 1)$, compute $\bar{f}$ and $\mathbf{p}'$.

3.4 Derive the n-type replicator equation from $N_i' = f_i N_i$. Explain the subtraction of mean fitness.

3.5 Verify (3.4) by differentiating its right-hand side. State the hypotheses that keep its exponent finite on a finite time interval.

3.6 Reduce M_s to the BC and CA edges. Determine the directions around the boundary cycle.

3.7 At $s = 1$, prove algebraically that the mean fitness is zero for every state. Is every state an equilibrium?

3.8 Verify $\bar{f} = (1 - s)Q$ and derive (3.6).

3.9 Prove the identity expressing $1 - 3Q$ as a sum of squared coordinate differences. Use it to determine its zero set.

3.10 Prove $p_A p_B p_C \leq 1/27$ for normalized nonnegative coordinates. Give both a calculus proof and an arithmetic-geometric-mean proof.

3.11 Compute the trace, determinant, and eigenvalues of (3.7) when $s = 1$. Explain what extra information the conserved product supplies.

3.12 Starting from the cyclic fitnesses, calculate all four entries of the reduced Jacobian at the center.

3.13 For $s = 1.3$, calculate the real and imaginary parts of the center eigenvalues. Interpret both quantities.

3.14 *Exploration.* Compare trajectories from $(0.55, 0.25, 0.20)$ at $s = 0.7, 1, 1.3$. Record the product at several times and relate its behavior to the direction of spiraling.

3.15 *Exploration.* Compare initial states $(0.4, 0.6, 0)$ and $(0.4, 0.59, 0.01)$ for $s = 1$. Explain why their long-term paths differ.

3.16 If all n fitnesses are equal at a state, prove the state is stationary. Prove the converse for an interior state. Explain how the converse changes on a face.

3.17 For an interior replicator trajectory, prove $\frac{d}{dt} \ln(p_i / p_j) = f_i - f_j$. Interpret the result for types with equal fitness along a trajectory.

3.18 *Exploration.* Implement a small-step numerical integration for the balanced cyclic model. Compare the error in $\sum_i p_i$ with the error in $p_A p_B p_C$. Explain why preserving one identity numerically need not preserve the other.

Selected Solutions

Chapter 3

3.1 The displayed position is $(0.65, 0.15\sqrt{3})$. The AB edge lacks C, the BC edge lacks A, and the CA edge lacks B.

3.3 $\bar{f} = 5/4$ and $\mathbf{p}' = (3/8, -5/16, -1/16)$. The three components sum to zero.

3.5 Differentiating the exponential gives $p_i' = p_i(f_i - \bar{f})$. Smooth fitnesses on a neighborhood of the compact simplex are bounded there, as is mean fitness, so their finite-time integrals are finite.

3.7 Antisymmetry gives $\mathbf{p}^T M_1 \mathbf{p} = 0$. Interior stationarity additionally requires $M_1 \mathbf{p} = 0$, which gives the uniform state. For example, at $(1/2, 1/3, 1/6)$ the velocity is nonzero.

3.9 Expanding the squared differences gives $\sum_i p_i^2 - Q$. Since $1 = \sum_i p_i^2 + 2Q$, this equals $1 - 3Q$. It vanishes exactly when all coordinates agree.

3.11 At $s = 1$, the trace is zero, determinant is $1/3$, and eigenvalues are $\pm i/\sqrt{3}$. Purely imaginary linearized eigenvalues alone do not establish nonlinear periodicity. The conserved product confines trajectories to smooth closed invariant curves.

3.13 The real part is 0.05, and the imaginary-part magnitude is $\sqrt{3}(2.3)/6 \approx 0.66395$. Small perturbations grow while circulating; their linearized period is about 9.46.

3.17 Subtract p_j'/p_j from p_i'/p_i. The two mean-fitness terms cancel. If the two fitnesses agree throughout a time interval, their probability ratio stays constant there.